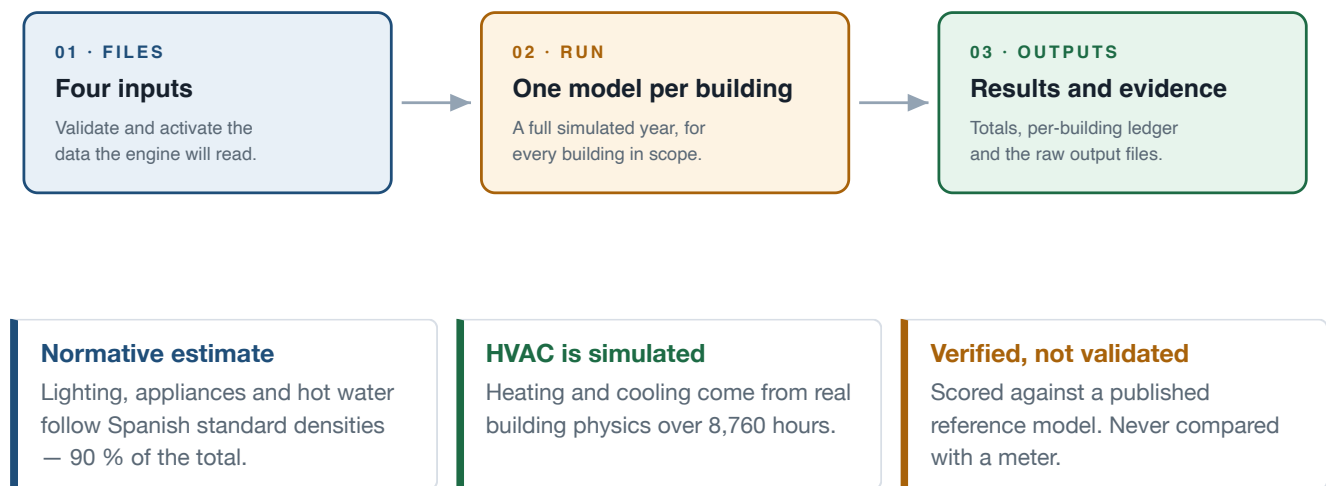


VISUAL GUIDE

Valencia Stock Energy Workbench

How the flow runs, which source supplies which value at every step, and how to read every number it produces.



Example run throughout

Benicalap district · `benicalap_v8`

Engine

EnergyPlus 25.2.0 · OpenStudio 3.11.0

Verified profile

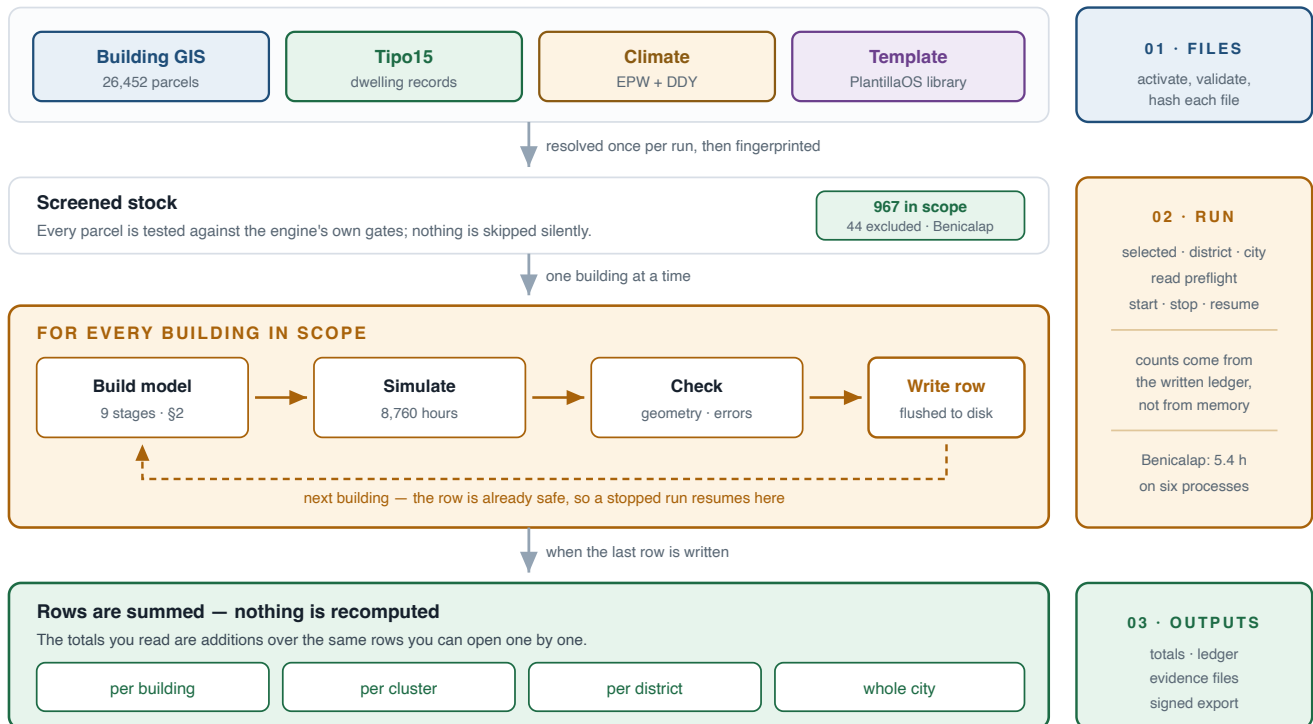
`80b3d4dafcdb7190...`

Installation

See `docs/getting-started.md`

1 The whole flow, on one page

Four files go in. Every building in scope gets its own model, its own annual simulation and its own row on disk. The rows are then added up. Nothing else happens.



There is no typological shortcut in this chain. A district result is the sum of that many individual annual simulations — not one representative building multiplied by a floor area.

2 Which source, which value, at every stage

This is what happens inside *Build model*. Each stage names the file it reads and the exact field it takes from it.

1

Footprint

The parcel polygon is simplified. If simplification moves its area too far, the building is excluded rather than distorted.
 GIS · polygon geometry, refparcela

2

Storey count — capped by what the cadastre records

Height sets the storeys, but they are limited to what the recorded dwelling area can fill, plus one commercial ground floor.
 GIS · altura_max → capped by Tipo15 · 442_sup_Residencial summed over 31_pc

3

Envelope, by typology and construction period

The building's TABULA class picks its walls, roof and floor — assembled layer by layer from real materials in the template.
 GIS · cluster (21 classes) → Template · named constructions and materials

4

Windows, frames and blinds

Discrete openings per façade with balcony doors, aluminium frames and Valencian *persianas* that close on strong summer sun.
 Template · glazing, frame and shading-control objects — placed on the geometry from stage 1

5

Neighbours — shading and party walls

Every real building within 50 m becomes shading mass; a shared wall becomes adiabatic instead of losing heat to the street.
 GIS · neighbouring polygons and their heights, read from the untouched cadastre

6

Occupancy — the people actually registered there

The registered resident count replaces the standard 20 m²/person norm, changing both internal gains and hygienic ventilation.
 GIS · pob_total, num_vivend → Template · CTE occupancy schedule kept unchanged

7

Domestic hot water

A hot-water loop sized on the Spanish CTE figure of 28 litres per person per day — so it follows the residents, not the floor area.
 GIS · pob_total × 28 l/day → Template · boiler and loop objects

8

Heat pumps, sized on design days

One packaged terminal heat pump per zone, autosized against the extreme winter and summer days rather than a rule of thumb.
 Climate · DDY winter + summer design days → Template · heat-pump performance curves

9

Annual simulation, then quality control

8,760 hours. The result is then checked back against the model's own geometry and the error file is scanned before the row is written.
 Climate · EPW hourly year → checked against stages 1–4, then a row on disk

Stage 2 is the one that changed most recently, and it is the difference between conditioning 2.76 and 3.84 million m² across this district. It is drawn on the next page.

3 The four inputs, and the gate each must pass

GEOMETRY + IDENTITY

Building GIS

The cadastre itself: one row per parcel, carrying the shape, the height, the typology class and the people.

FIELDS THE ENGINE READS

refparcela · altura_max · cluster
pob_total · num_vivend · nombre · geometry

GATE Required fields present, geometry readable, coordinate system recorded.

RESIDENTIAL AREA + USE

Tipo15 companion

The cadastre's floor-by-floor record: how much dwelling area exists, on which floor, for what use.

FIELDS THE ENGINE READS

31_pc (joins to refparcela)
252_planta · 428_uso · 442_sup_Residencial

GATE All four columns parseable; the join to parcel references is measured and reported, not assumed.

ANNUAL + SIZING WEATHER

Climate pair

Two files that must agree with each other: the year that is simulated, and the extremes the equipment is sized on.

WHAT EACH ONE DRIVES

· epw — every one of the 8,760 simulated hours
· ddy — heat-pump capacity in stage 8

GATE A complete 8,760-hour year; a winter and a summer design day. The two are cross-checked as one pair.

OPENSTUDIO SOURCE

PlantillaOS template

The library every model is assembled from. It carries no geometry — the geometry is built per building.

WHAT IT SUPPLIES

constructions and materials · schedules · space types
equipment definitions · thermostat set points

GATE Every construction, schedule and space type the engine will ask for must already exist — and no geometry.

Nothing is hard-coded into the program. A newer cadastre, a different weather year or a modified template can be swapped in from this screen without touching any code — provided it passes its gate.

Why every file is hashed

Each file is copied into managed storage and its checksum recorded. A run resolves the active set once and stamps those checksums into every result row — so any row can later be traced to the exact bytes that produced it.

4 The two things that move the number most

Stage 2 — a parcel is not always one building

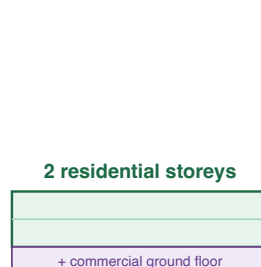
PARCEL 3748901YJ2734H · BENICALAP · BlocPluriP06 · altura_max = 15

Before — extruded to full height



17,273 m² footprint × 16
276,366 m² conditioned
5.6 % of the whole district's energy

After — capped by the recorded dwellings



342 dwellings · 38,158 m² recorded
51,819 m² conditioned
1.6 % of the district's energy

Across Benicalap

399 of 967 buildings were capped.
They carry 55 % of the district's energy after the correction.

521 buildings came out identical to the bit — they were already short enough.

District total: 138.15 → 109.76 GWh

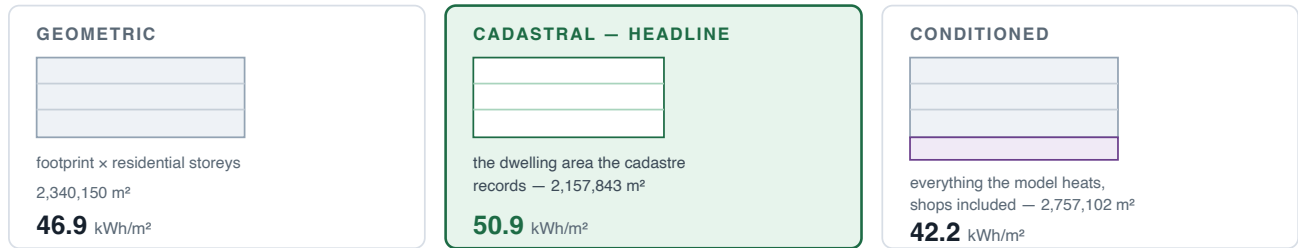
The footprint is the whole city block; the height is that of its tallest corner. Multiplying them invents floor area that does not exist, so the storeys are limited to what the cadastre can actually account for.

What the cap does not fix

A capped building gets the right area but keeps the full block footprint, so it is squat where the real building is a tower and its surface-to-volume ratio is wrong. Because heating and cooling are only 8.6 % of the total, the area correction dominates and the shape error is second-order — but it is real, and it stays on the list in §7.

Which floor area is the number divided by?

The same **109.76 GWh** divided three different ways. Every intensity in this product states which one it used.

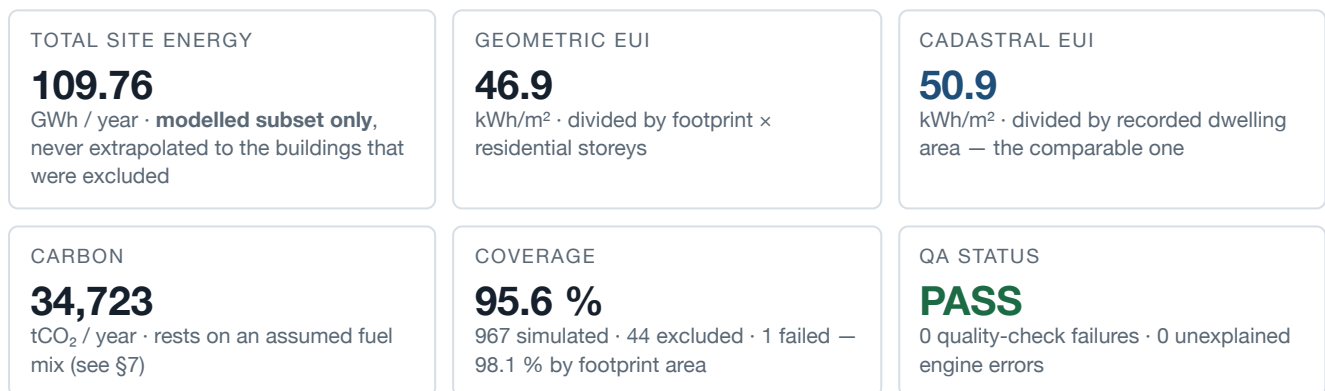


▲ Same energy, three answers, a 20 % spread. A comparison that does not state the basis is not a comparison.

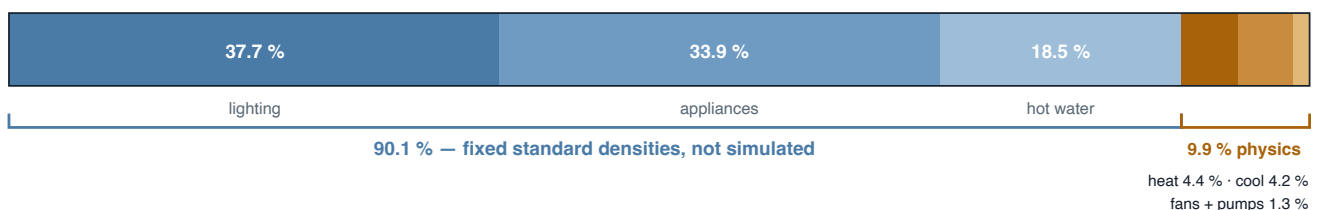
The headline uses the cadastral basis because that is the basis the published reference study uses — which is what makes the “Δ vs Rai” column on the Outputs screen a like-for-like comparison.

5 Reading the Outputs screen

These are the six figures at the top of the screen, filled in with the Benicalap run. The small print under each one is the basis — it is there because it changes what the number means.



Where the energy actually comes from



This is the single most important fact about the result. Only heating, cooling, fans and pumps come out of building physics. Work on insulation, shading or thermal bridges therefore moves under a tenth of the headline — which does not make it worthless, but does tell you what the headline is sensitive to.

BELOW THE KPIS	WHAT IT GIVES YOU
Cluster totals	One row per TABULA class, with <i>both</i> denominators side by side, the reference study's published intensity ("Rai reference") and the percentage difference against it.
Building ledger	Every building as its own searchable row: status, cluster, intensity, energy, CO ₂ , occupancy plausibility, QA verdict. Filterable by <code>ok</code> , <code>failed</code> or <code>excluded</code> .
Preserved files	Opening one building gives five files kept from its own run: <code>eplustbl.htm</code> (the EnergyPlus results table), <code>deep_layers.json</code> (what each stage did to this building, plus its QA checks), <code>model_python.osm</code> , <code>eplusout.err</code> and <code>verified_profile.json</code> .
Export	The building table as CSV, or a signed package — one building or the whole run — with an Ed25519 manifest.

6 How to check that we are not fooling ourselves

The tool rebuilds a building published by the reference author — from the surface areas he reported — and scores itself against his own results on thirteen quantities. Any failure makes the command exit non-zero.

QUANTITY	OURS	REFERENCE	DIFFERENCE	TOLERANCE
Residential floor area (m ²)	954.800	954.800	0.00 %	0.5 %
Wall U-value (W/m ² K)	1.409	1.409	0.00 %	1 %
Lighting (GJ)	65.308	65.320	-0.02 %	1 %
Hot water (GJ)	36.057	36.080	-0.06 %	1 %
HVAC sensible heating (GJ)	26.527	27.300	-2.83 %	5 %
HVAC sensible cooling (GJ)	-68.403	-65.030	+5.19 %	8 %
Total site energy (kWh/m²)	53.680	54.500	-1.50 %	3 %

Read this honestly

The reconstruction shares the reference author's template and weather file and is built from the surface areas he reported. It verifies the *engine and the parameter set*. It does not independently confirm that the typological abstraction, or the wider stock geometry, is right.

- **A profile fingerprint.** Every locked parameter and the checksum of every engine source file are hashed into one identifier (`80b3d4dafcdb7190...`) written into every result row. When the engine is later corrected the fingerprint is retired: the old result stays valid and traceable, but reproducing it byte-for-byte means checking out the source state it points to, not the newest version.
- **A drift guard.** If any engine source changes, a run refuses to start and names the parameter that moved — so no single run can span two versions of the physics.
- **Self-describing rows.** Each row carries the fingerprints of the inputs, the policy and the engine that produced it, so one row can be traced without the rest of the run.

7 Where this stands — my own notes

I would rather you heard this from me than found it yourself. Everything below is measured, not estimated, and the sizes are from the Benicalap run.

A What I cannot do today

- **I cannot tell you what a building actually consumes.** Nothing here has been compared with a meter. I report what the standards and the physics imply — not what happened. Affects every number in the product.
- **I cannot give you an uncertainty band.** The sampling study exists, but it was run on an earlier configuration of the engine, so the current figures carry no confidence interval. A range would have to be re-derived before I could quote one.
- **I cannot split heating and cooling between the dwellings and the shops.** EnergyPlus meters by end use, not by space type, so the residential subtotal I report is an *upper bound* rather than a clean split. Commercial ground floors are 10.0 % of site energy.
- **I cannot justify the carbon figure.** It rests on an assumed fuel mix for the real stock that nobody has confirmed for me yet. 34,723 tCO₂/yr — treat it as conditional.
- **I cannot recover the true shape of a capped block.** I get its area right and its massing only approximately; it stays squat where the real building is a tower. 399 buildings · 55 % of district energy.
- **I cannot resolve a deep floor plate properly.** Above a footprint threshold each storey is one well-mixed zone, with no core-versus-perimeter distinction. 9 buildings · 9.2 % of area · 8.0 % of energy — flagged per building, never silent.
- **I cannot model more than one weather year**, and I cannot model a warmed future climate. One measured year is used throughout. A single EPW file.
- **My totals cover the modelled subset only, and that subset is not a neutral sample.** Exclusions concentrate in large buildings, and larger footprints tend to have lower intensity — so the intensity I report is biased slightly upward relative to the full stock. 95.6 % of buildings · 98.1 % of footprint area.

B What could be better, but is not blocking

- **Lighting and appliances follow a fixed standard density.** That is 71.6 % of the site energy coming from a norm rather than a measurement. I keep it because the reference study uses the same assumption, which is what makes the comparison consistent — but consistency is not accuracy. No appliance survey exists for this stock.
- **The smallest model I can build is ground plus one storey**, so very short buildings cannot shrink any further even when the cadastre says they should. 74 buildings still condition more than twice their recorded dwelling area.
- **A building with no registered residents gets no hot water at all.** This is deliberate and matches the reference thesis; the alternative policy is implemented and can be selected instead. Policy is recorded in every row.
- **Some buildings have implausible population records** — far too many or too few people for their floor area. I flag them per building rather than correcting them, because the register is the source and I would rather you saw the anomaly than a quiet guess. 17 of 967 flagged in this district.
- **The envelope data covers only part of the stock.** The published reference models span roughly 31 % of Valencia's buildings; for the largest family — around 18,000 blocks, some 69 % of the stock — I am working from typological tables rather than a published model. I have asked Rai for the missing envelope definitions; that request is what closes this item.

C What I am working on right now

- **The full-city run.** The engine already screens 25,094 of the 26,452 parcels as buildable. I am deliberately not launching it until the envelope question above is settled — a five-day run on an assumption I expect to change is a waste of five days.
- **Extending the envelope data**, which is the item that gates the full-city run.
- **Transferring the method to a second city.** A full-stock run on Lecco completed and verified itself internally, but it has no external validation, so I label it a transferability scenario rather than a result.

D What comes next, in order

1. **Settle the envelope question, then run the whole city.** Everything else is smaller than this.
2. **Put an uncertainty band on the numbers** by re-running the sampling study against the current configuration. Until then every figure is a point estimate with no stated spread.
3. **Separate HVAC by space type** so the residential subtotal becomes a real split rather than an upper bound. This needs an extra output table and a re-run.
4. **Calibrate against measured consumption.** This is the one that would change the standing of everything else — until measured data arrives, no number here can be presented as real consumption, only as what the standards and the physics imply.

If you want to go past the three tabs

- **One building from the command line.**

`python src/verified_model.py --refparcela <REFERENCE>` runs the identical pipeline for a single building. `--verify` re-runs the thirteen checks in §6; `--show-profile` prints every locked parameter.

- **Earlier research screens.** A geometry builder, a full OpenStudio editor, scenario controls, the representative-cluster pipelines and the sampling study are all still in the application, hidden from navigation but reachable by direct URL. They belong to the earlier method and are kept for reproducing historical results, not for new work.

TERM	MEANING
<code>refparcela</code>	The cadastral reference identifying a parcel. The primary key everywhere in this tool.
Cluster	A TABULA class combining building type with construction period P01–P07. Twenty-one exist in Valencia.
EUI	Energy Use Intensity — annual energy over floor area, in kWh/m². Meaningless unless the basis is stated (§4).
Tipo15	The cadastral companion file listing dwellings floor by floor: how many, what use, how much area.
<code>pob_total</code>	Registered residents per building, from the municipal register — actual residence, not design occupancy.
EPW / DDY	The annual hourly weather file, and the extreme design days used to size equipment.